

INSTITUT UNIVERSITAIRE DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIE (FST)

RAPPORT DE TRAVAUX DIRIGÉS IV

Cours : Réseaux I

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Enseignant : SAINT AMOUR Ismael

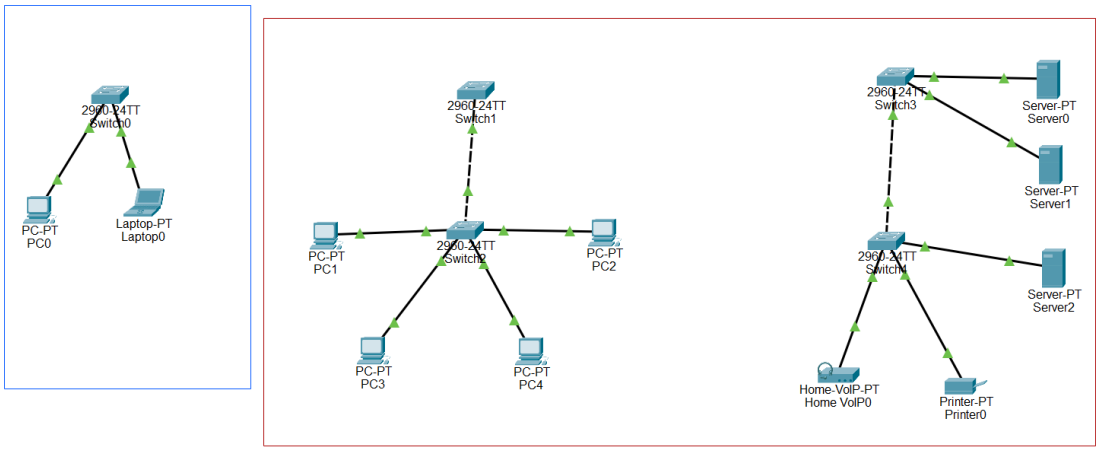
Niveau : L3

Date : 11 novembre 2026

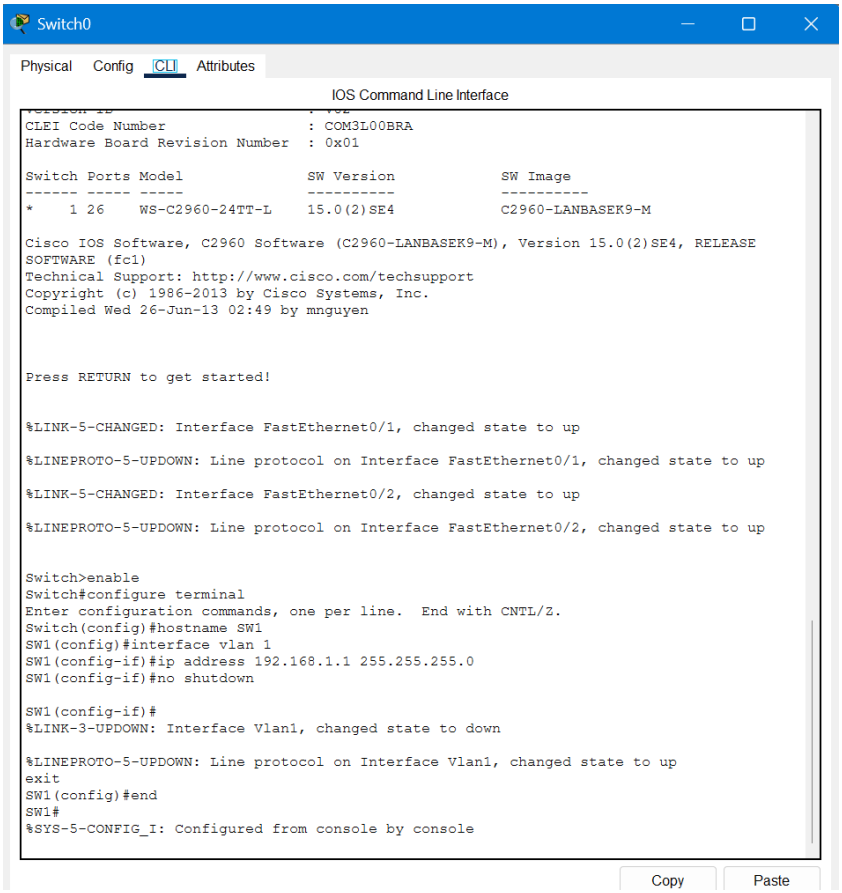
Dans la première partie du TD on avait à reproduire une topologie avec les composant qui suivent :

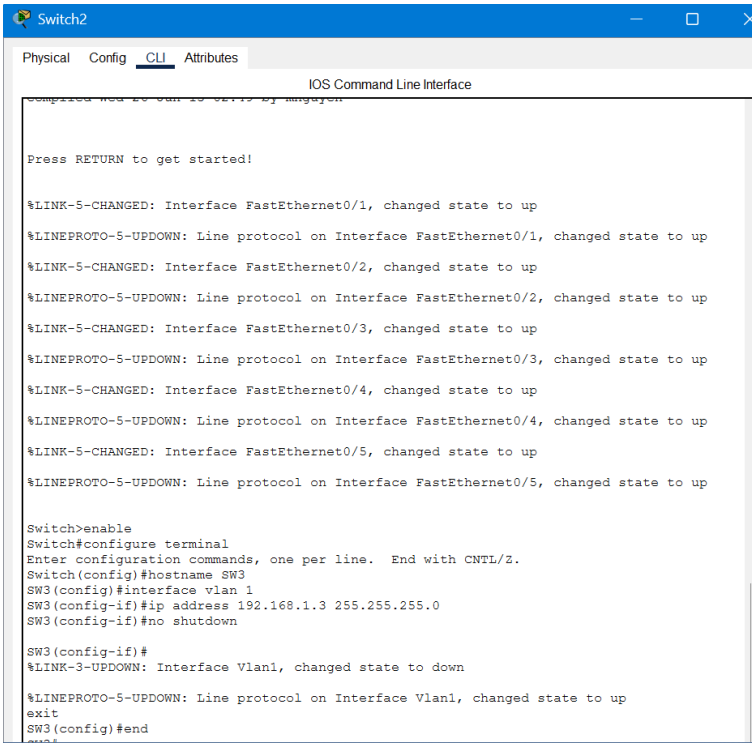
5 switch, 5 PC, une laptop, 3 server un printer et un Home-VoIP-PT phone. Le but était de créer deux réseaux et de configurer des adresses Ipv4 ou Ipv6 et tester la connectivite via la commande Ping et du mode de simulation.

Dans l'image qui va suivre on va illustrer la topologie

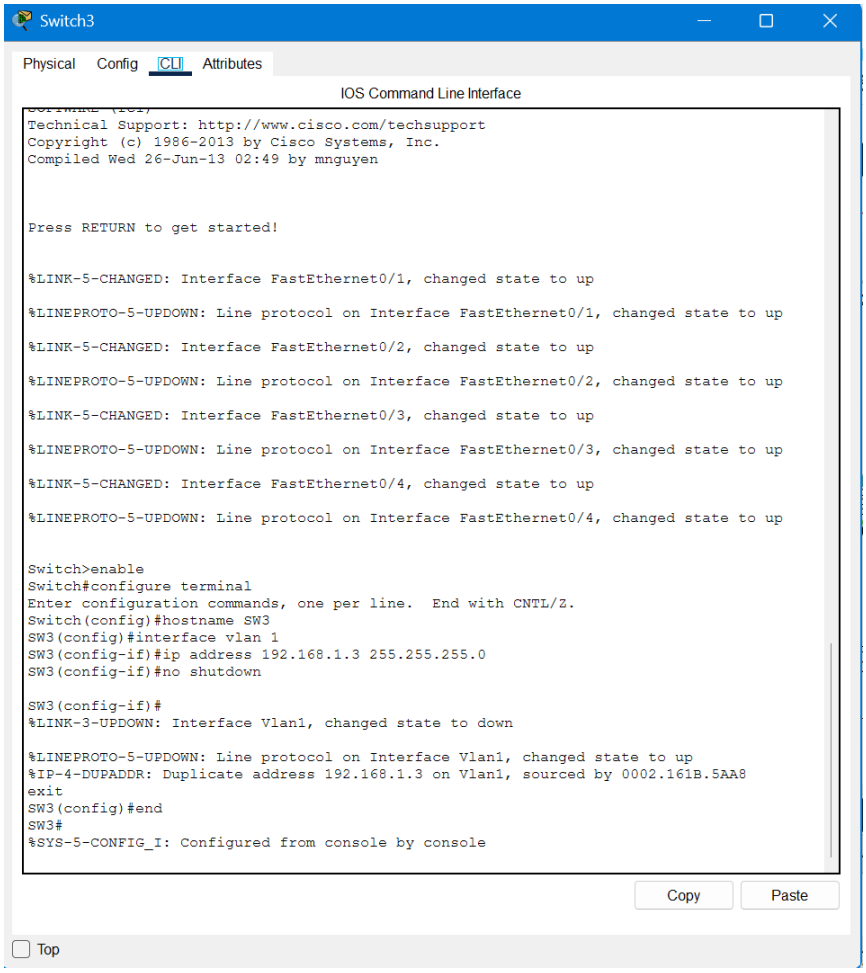


Avant de procéder au test, on doit configurer chaque switch

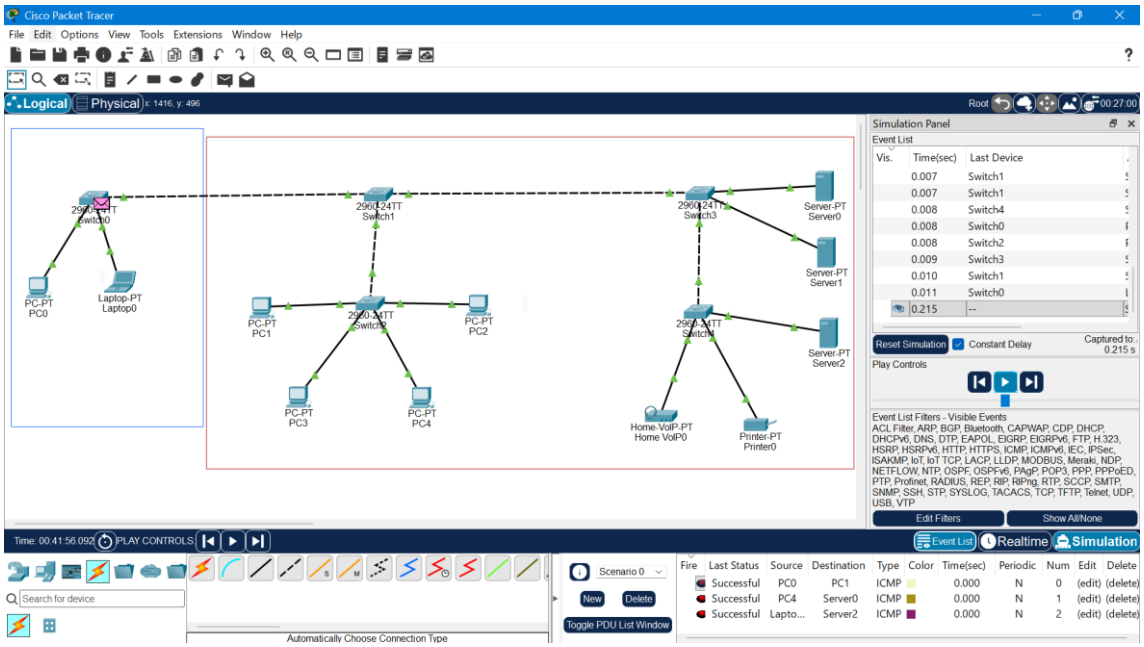



Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#interface vlan 1
SW3(config-if)#ip address 192.168.1.3 255.255.255.0
SW3(config-if)#no shutdown
SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
SW3(config)#end

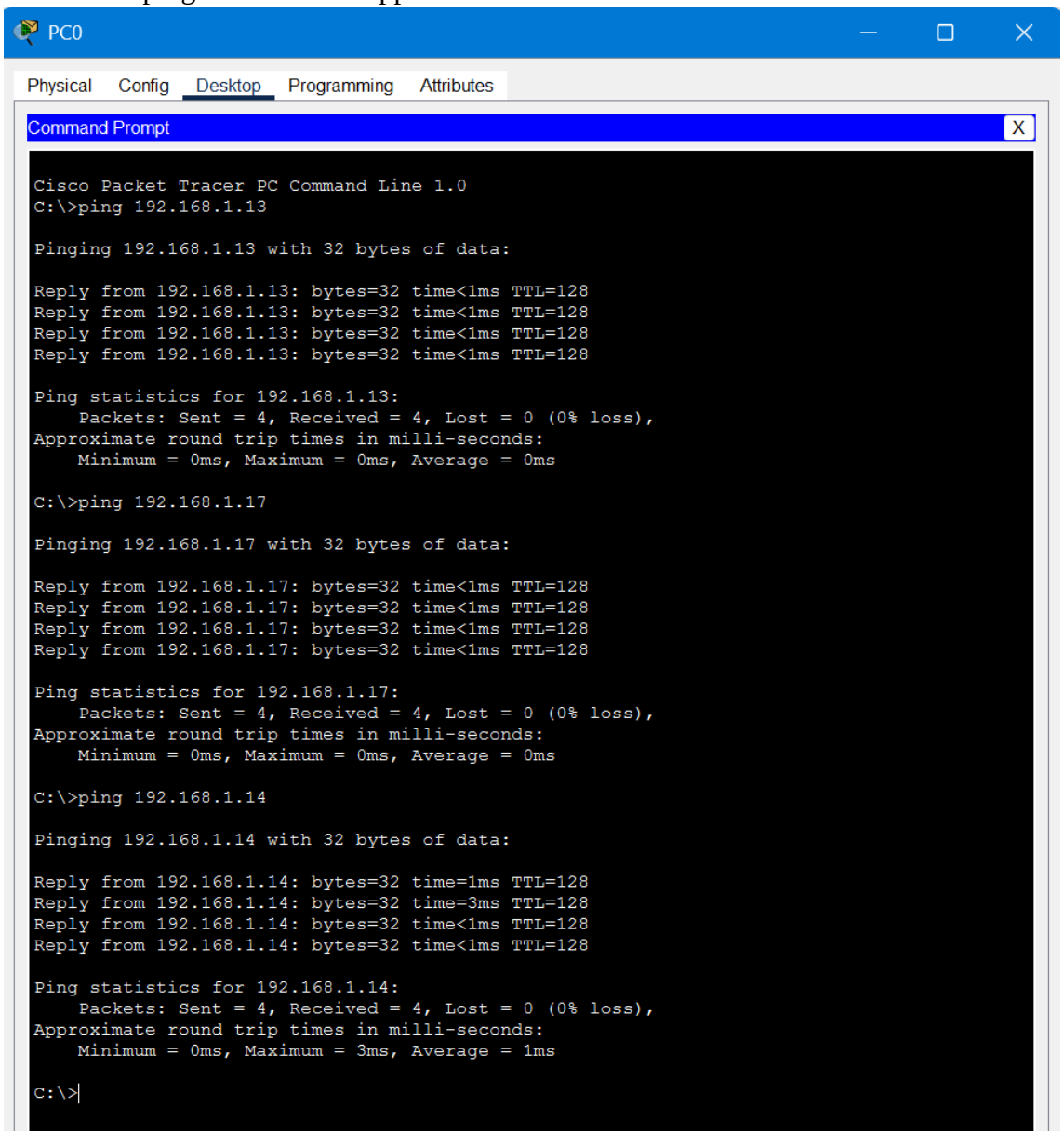

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.1.2 255.255.255.0
SW2(config-if)#no shutdown
SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
Copy
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Une fois configurer, on a effectué d'abord un test de simulation

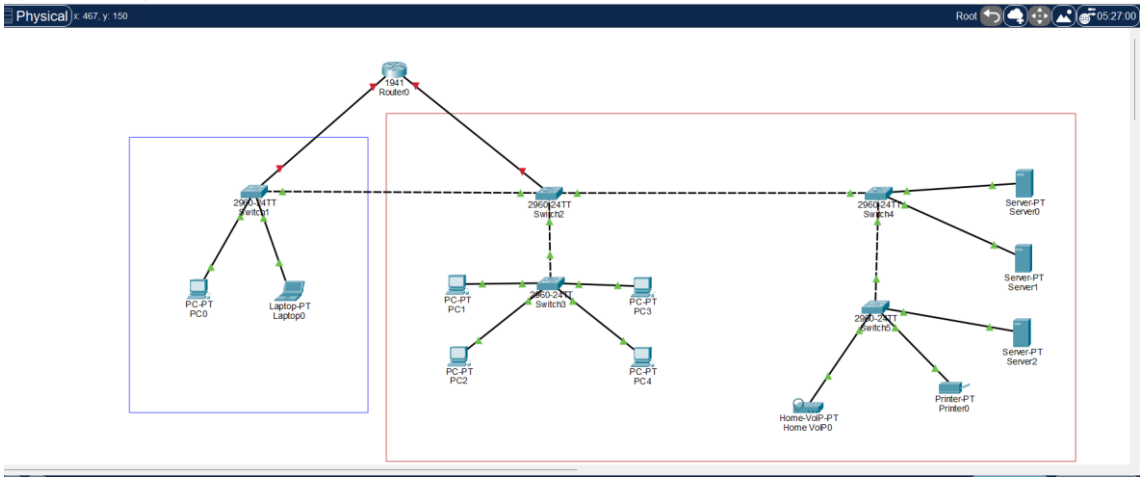


Et aussi un ping sur les autres appareils via le PC0



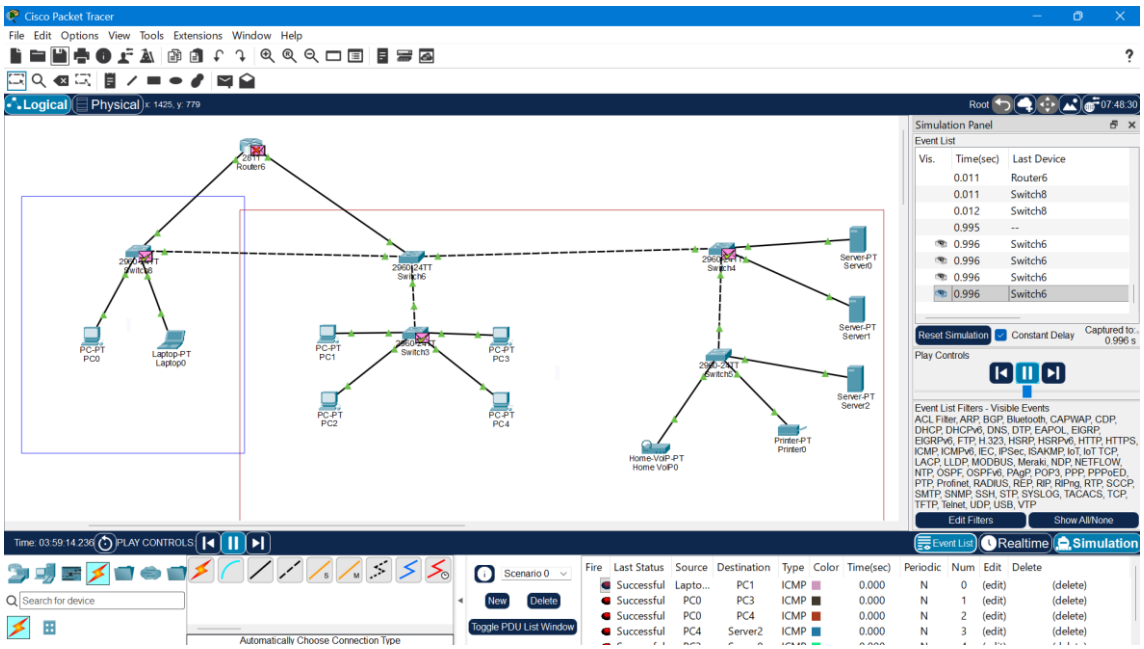
La connectivité s’est montrée positive.

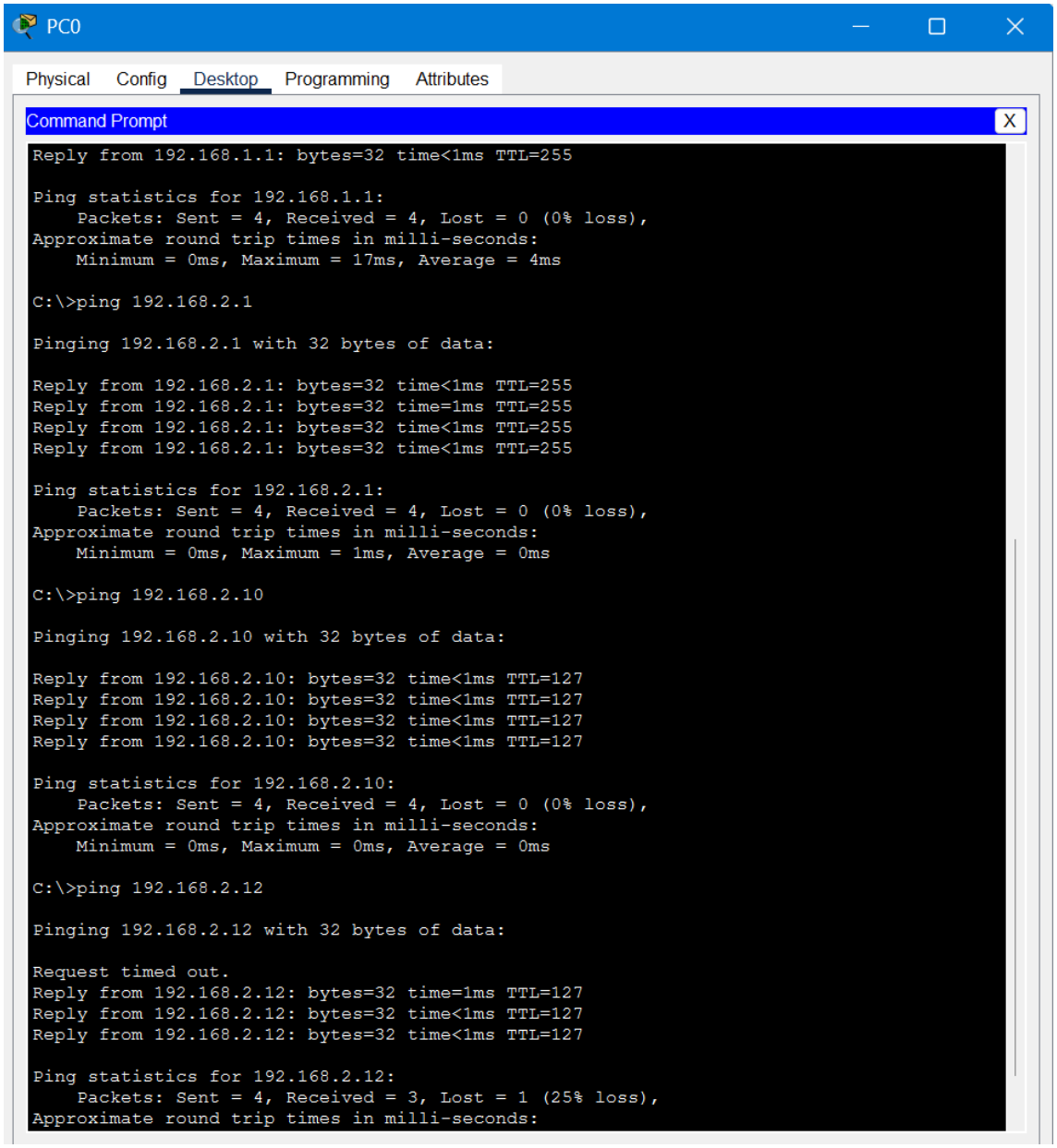
Dans la deuxième partie on va ajouter un routeur a notre réseau, qui va leur permettre a nos deux réseaux de pouvoir communiquer



On a configuré les deux interfaces du router FastEthernet 0/0 avec l'adresse 192.168.1.1 et FastEthernet0/1 avec 192.168.2.1

Voici le test de simulation




PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 4ms
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.12
Pinging 192.168.2.12 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.12: bytes=32 time=1ms TTL=127
Reply from 192.168.2.12: bytes=32 time<1ms TTL=127
Reply from 192.168.2.12: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.2.12:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:

Conclusion

En somme cette expérience nous a permis de bien comprendre comment deux réseaux peuvent réussir à communiquer via les passerelles qu’on a configurées sur chaque hôte.

Pour réussir cette évènement on a dû configurer les deux interfaces de notre routeur, avec une adressage d’IP différente. Sans Passerelles il est strictement impossible que deux réseau distinctes puisse effectuer des communications.

Les premières simulations ce sont solde par des échecs, car malgré le fait que j’ai fait un bon adressage sur les appareils je n’avais pas utilisé de passerelle. Donc j’ai assigné des passerelles aux différents appareils et ils ont fini par échanger des données.

La tâche est pleinement réussie